第一章 几何光学

第一节 几何光学基本定律

§1.1 几何光学三定律

§1.2 全反射

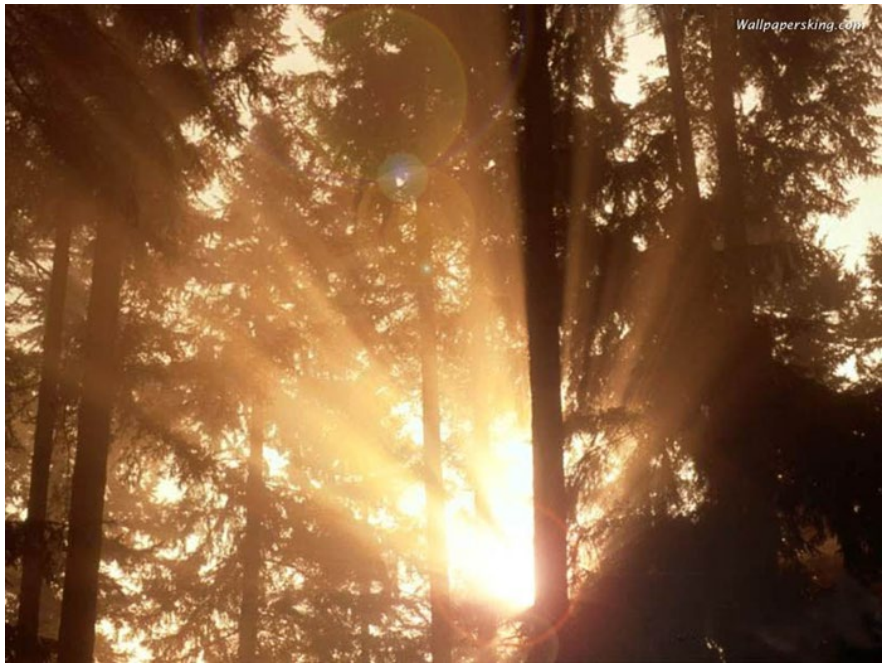
§1.3 棱镜和色散

§1.4 光的可逆性

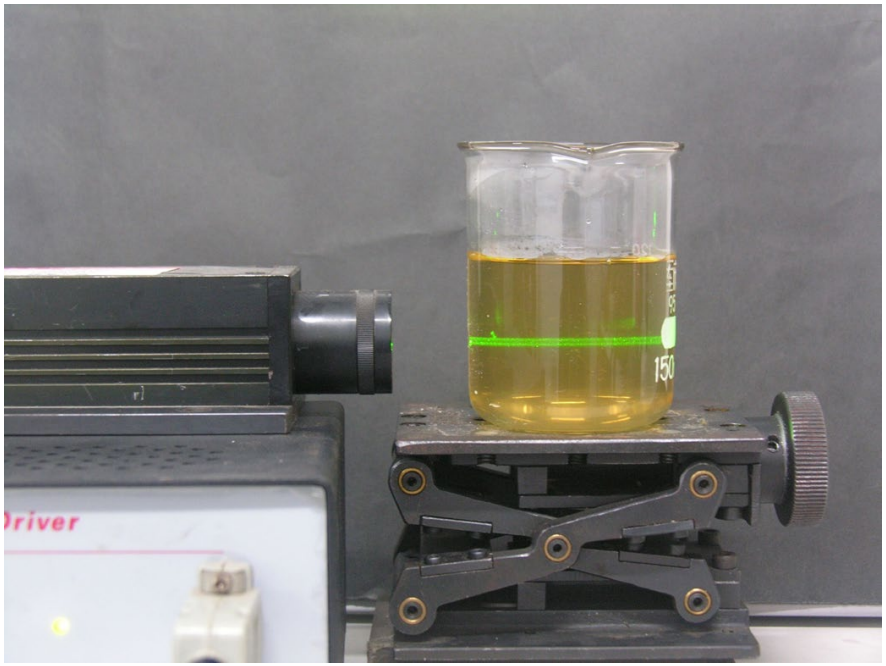
几何光学：光的直线传播、反射/折射定律为基础，研究光在介质中的传输规律及其应用。

§1.1 几何光学三定律

① 光的直线传播定律：光在均匀介质里沿直线传播



光线的概念



激光束在液体中的直线传播

§1.1 几何光学三定律

① 光的直线传播定律：光在均匀介质里沿直线传播

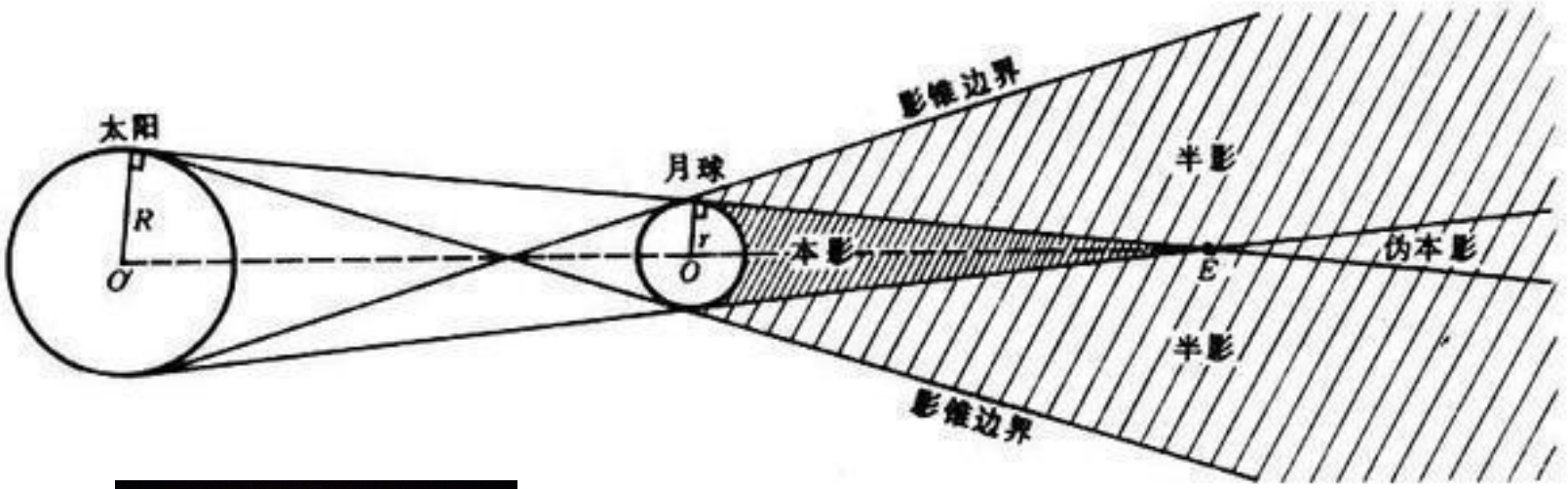
Chinese shadow puppetry

影的形成：皮影戏

墨经：小孔成像

§1.1 几何光学三定律

① 光的直线传播定律：光在均匀介质里沿直线传播

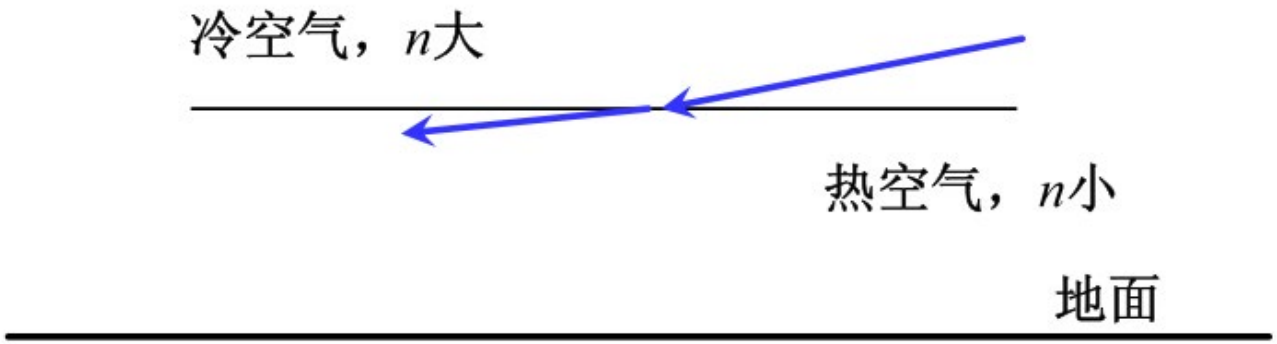


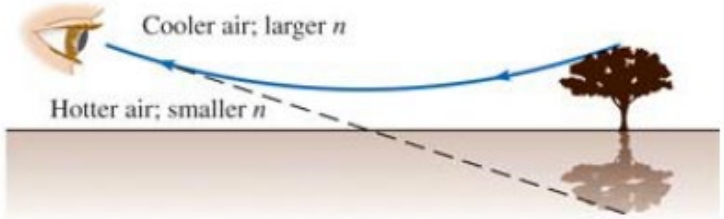
日食、月食等天象

有没有光不沿直线传播的情况？

1. 在非均匀介质中

例：沙洲神泉





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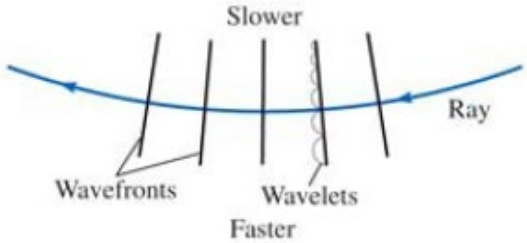
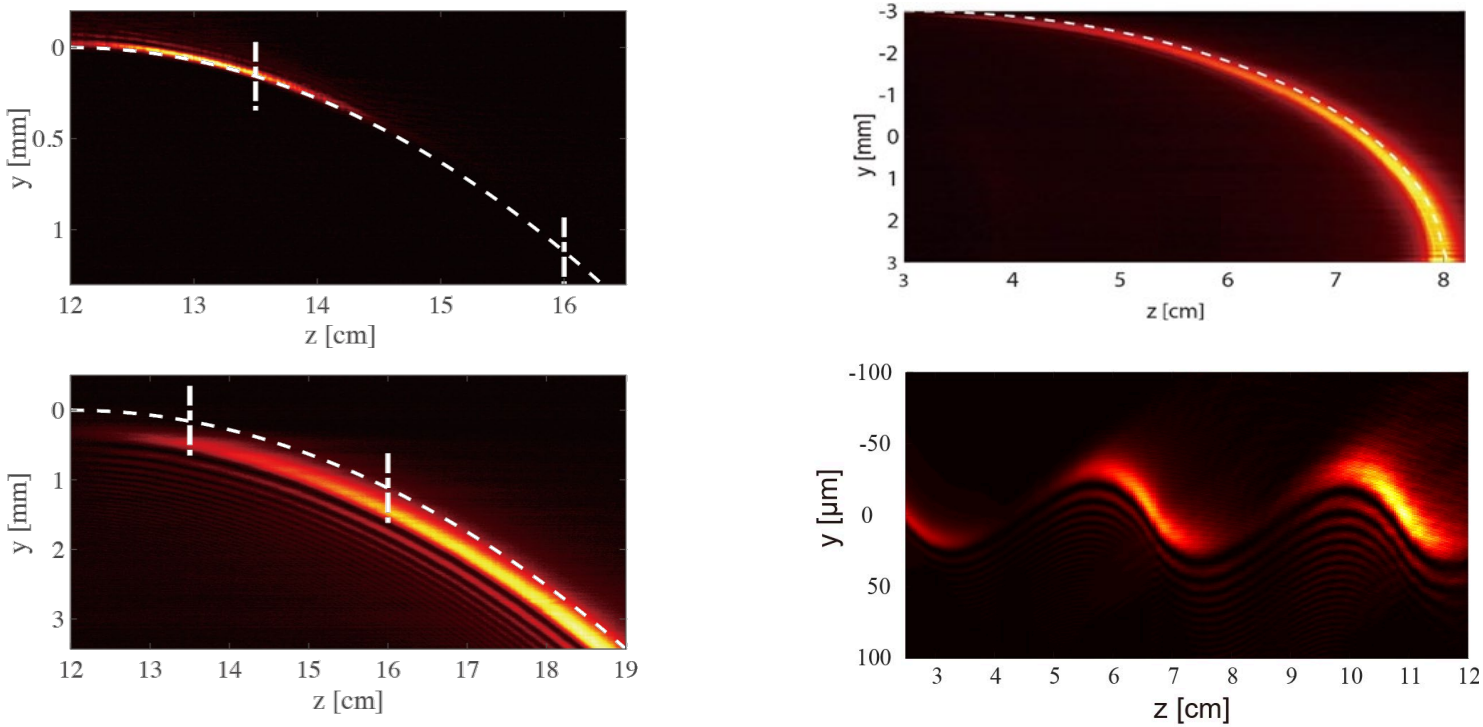


有没有光不沿直线传播的情况？

2. 在均匀介质中　　特殊光束：艾里光束等弯曲光束

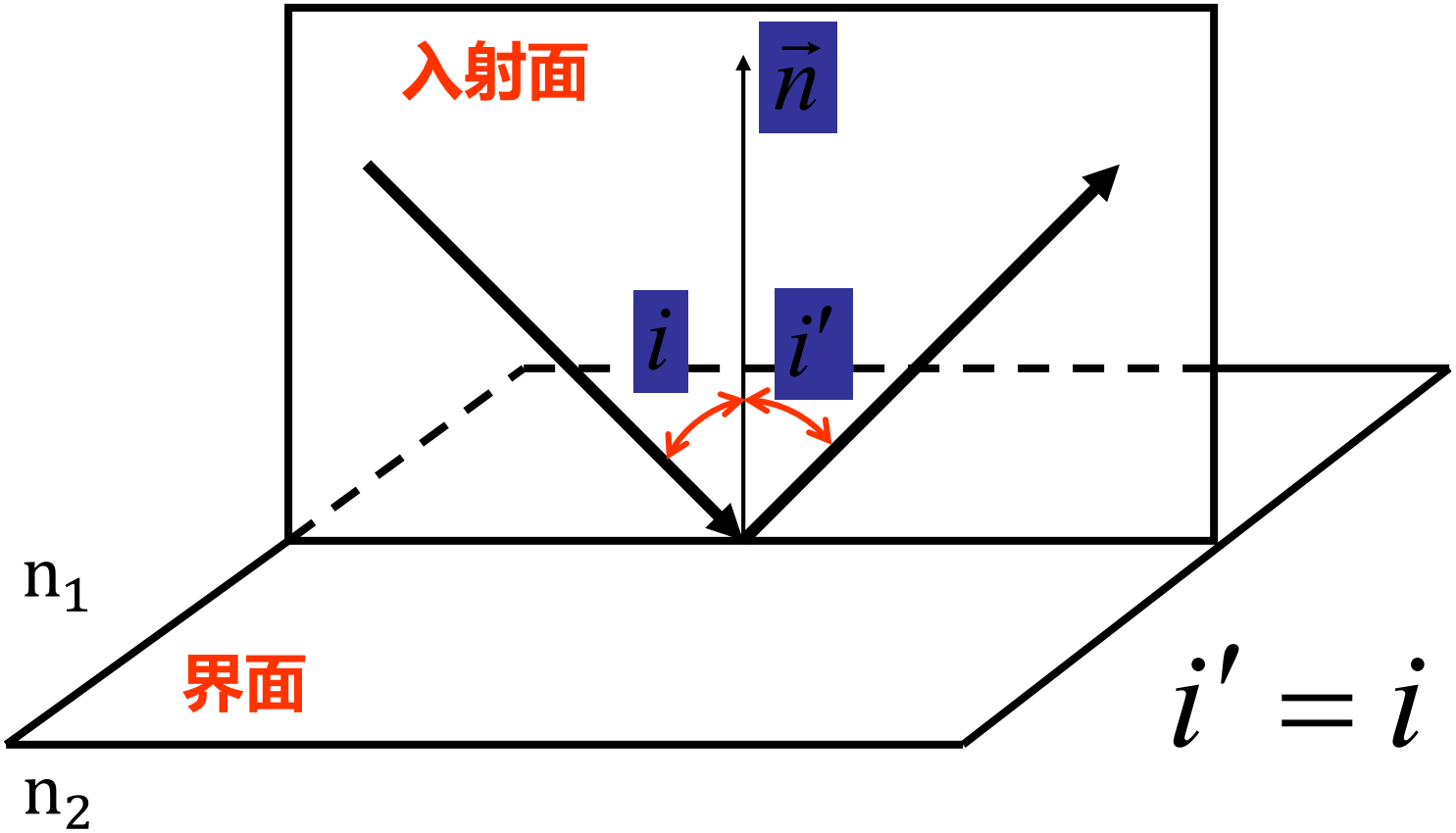


3. 在非线性介质体系中通过非线性相互作用实现

§1.1 几何光学三定律

② 光的反射定律

— 反射光线在入射面内
— 反射角等于入射角



$$i' = i$$

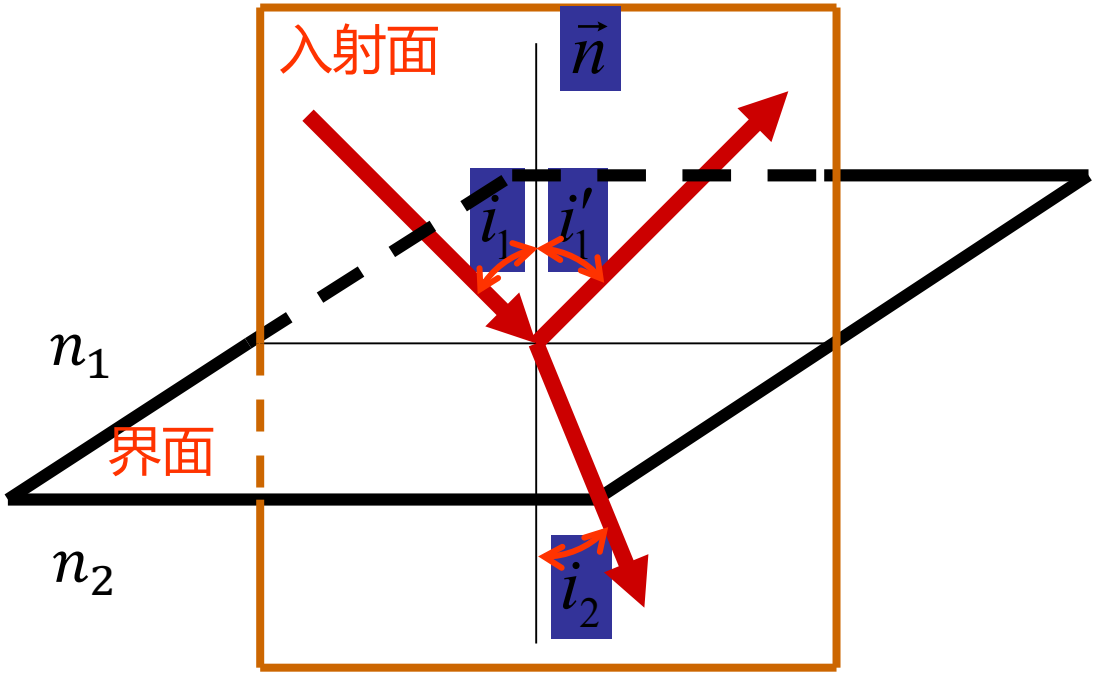
§1.1 几何光学三定律

③ 光的折射定律

1）折射光在入射面内

2）折射角、入射角正弦之比等于相对折射率

$$\frac{sini_1}{sini_2} = n_{12} = \frac{n_2}{n_1}$$



斯涅尔（Snell）定律

$$n_1 \sin i_1 = n_2 \sin i_2$$

例题1 水中的光点：

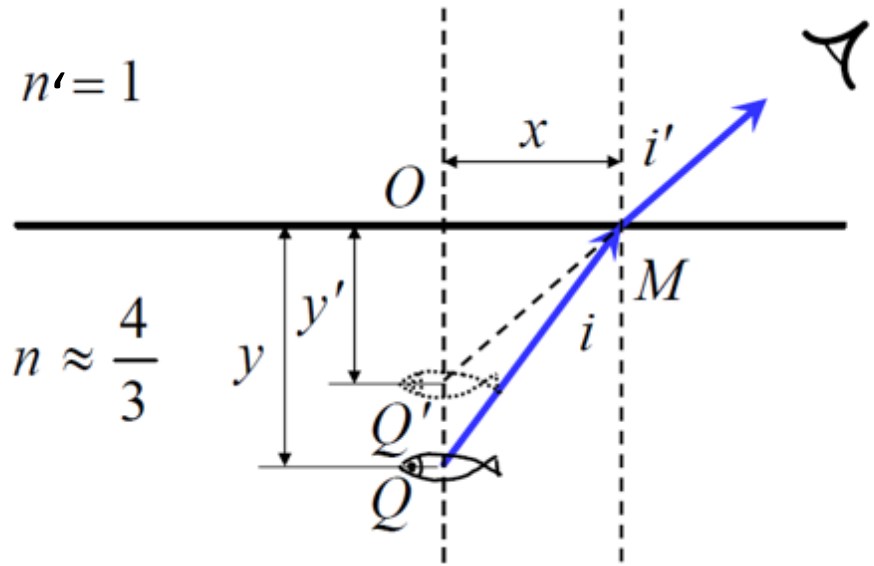
在水中深度为y处有一发光点Q，作QO垂直于水面，求射出水面的延长线与QO交点Q'的深度y'与入射角的关系。

解： 根据折射定律有

$$nsini = sini'$$

设入射点M到O点的距离为x

$$tani = \frac{x}{y} \qquad tani' = \frac{x}{y'}$$



消去x，则得

$$y' = y\frac{tani}{tani'} = y\frac{sinicosi'}{sini'cosi}$$

$$= \frac{y\sqrt{1 - n^2 sin^2 i}}{ncosi}$$

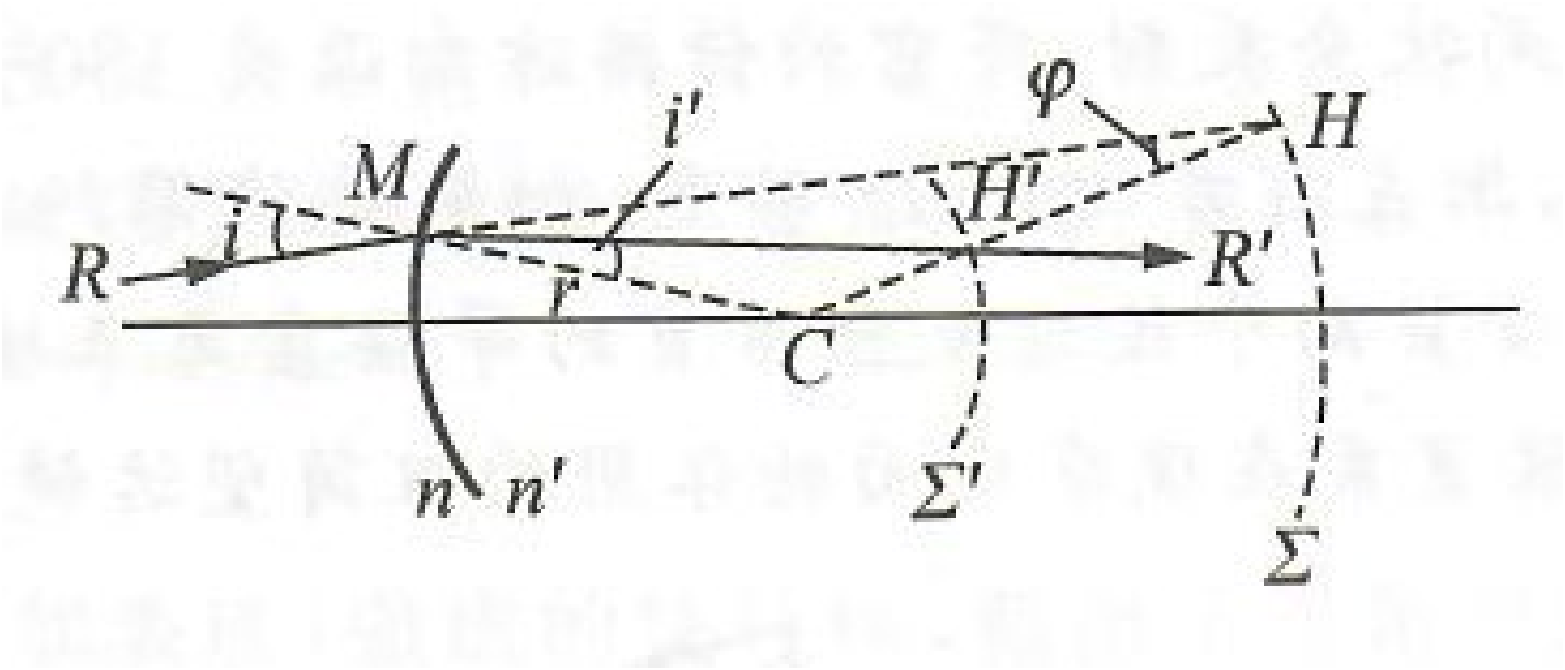
若在垂直方向观测（$i \approx 0$），则有

$$y' = \frac{y}{n} = \frac{3}{4}y$$

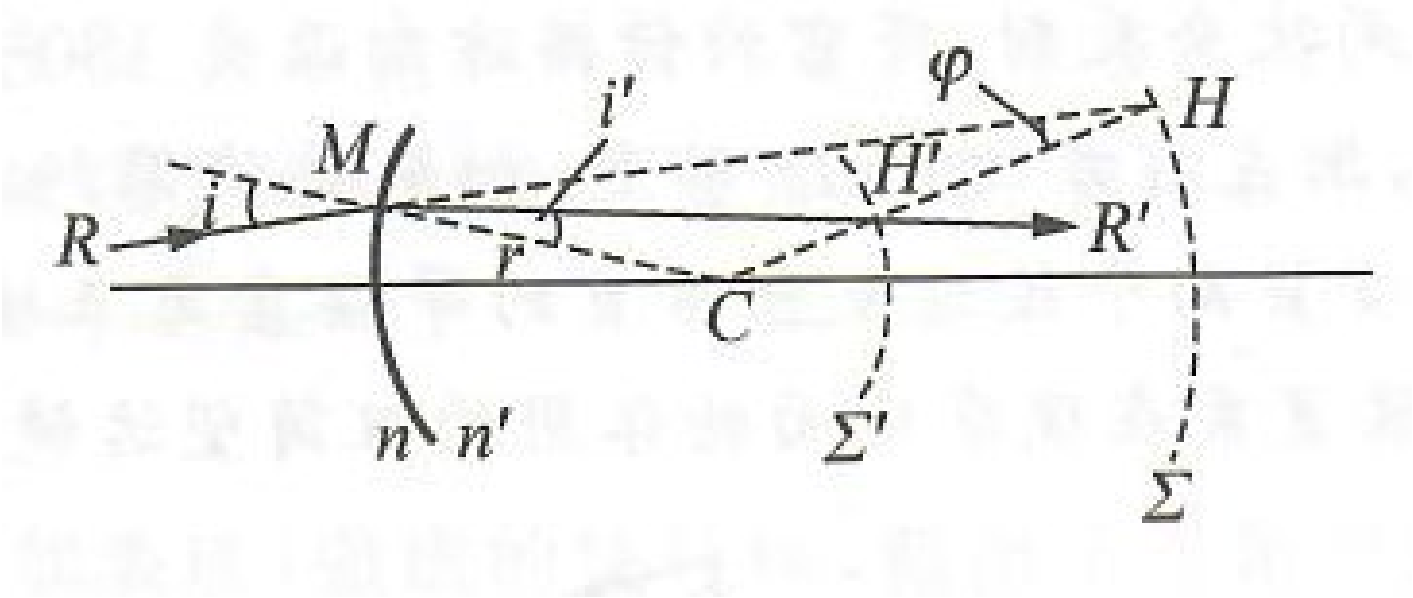
例题2　用作图法求任意入射线在球面上的折射线

假设折射球面的球心在C点，球面半径为r。设 $n < n'$

以C为中心，分别以$\rho = \frac{n'}{n} r$和$\rho' = \frac{n}{n'} r$为半径作两个圆弧面E和E’，然后将入射线RM延长，与圆弧面E交于H，连接CH，交圆弧面E’于H’，则MH’就是折射线。



例题2　用作图法求任意入射线在球面上的折射线



说明：对于三角形ΔHCM　　对于$\Delta MCH'$和ΔHCM，有公共角$\angle C$

$$\frac{sini}{sin\varphi}=\frac{CH}{CM}=\frac{n'}{n}$$

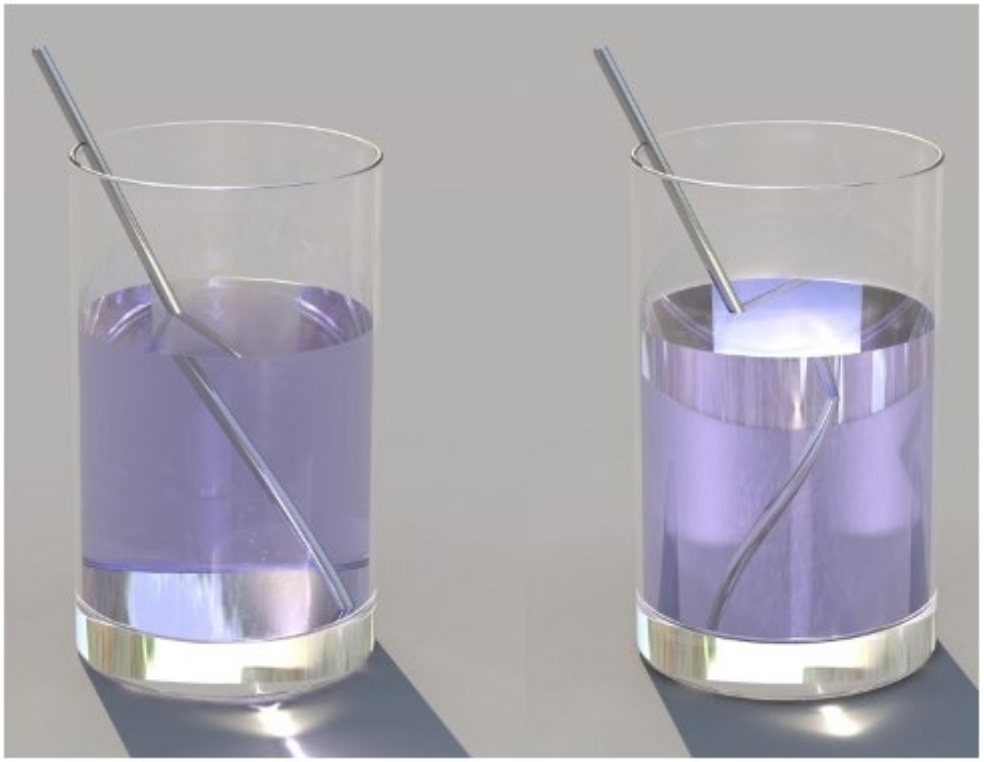
且 $\frac{CH}{CM}=\frac{CM}{CH'}=\frac{n'}{n}$　故 $i'=\varphi$

因此　$nsini=n'sini'$

一些奇怪的折射效应

Do you believe that n may be smaller than 1.0?

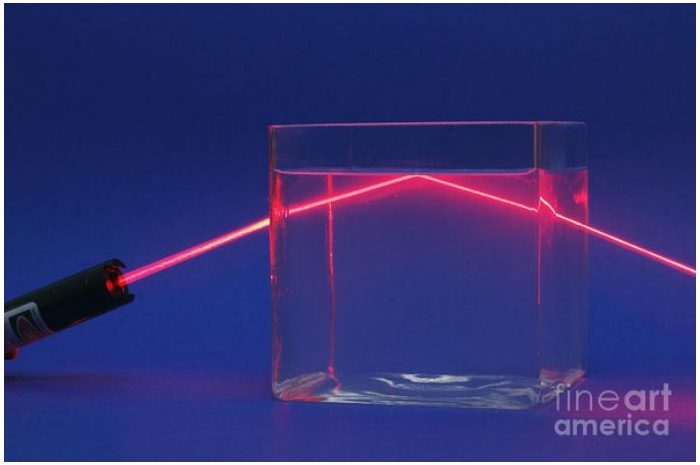
Do you believe that n is negative?



§1.2 全反射

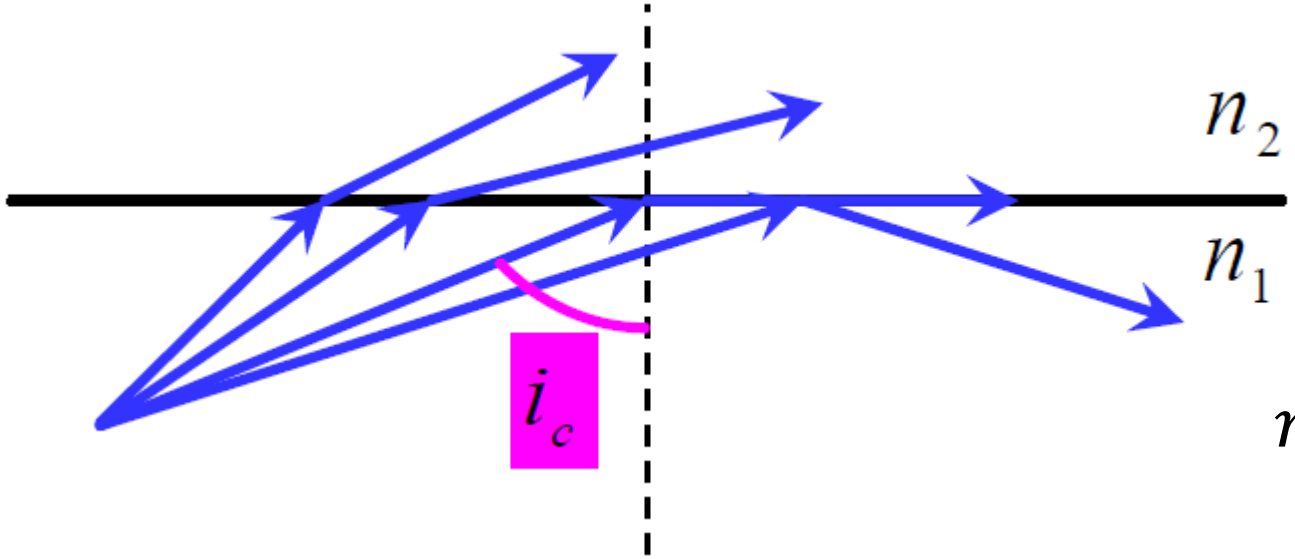
光密介质向光疏介质的折射：

$$\sin i_1 = \frac{n_2}{n_1}\sin i_2$$



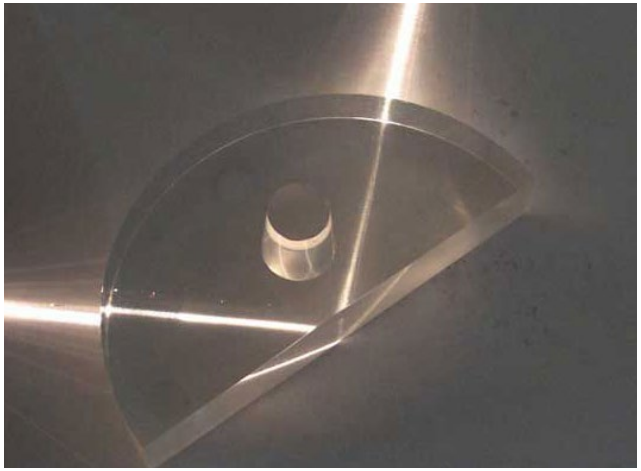
临界角（critical angle）：

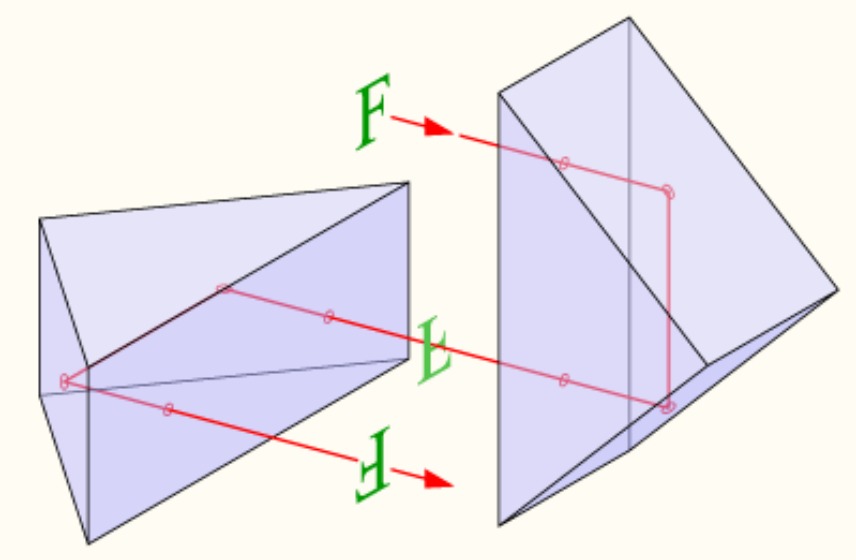
$$i_c = \arcsin\frac{n_2}{n_1}$$

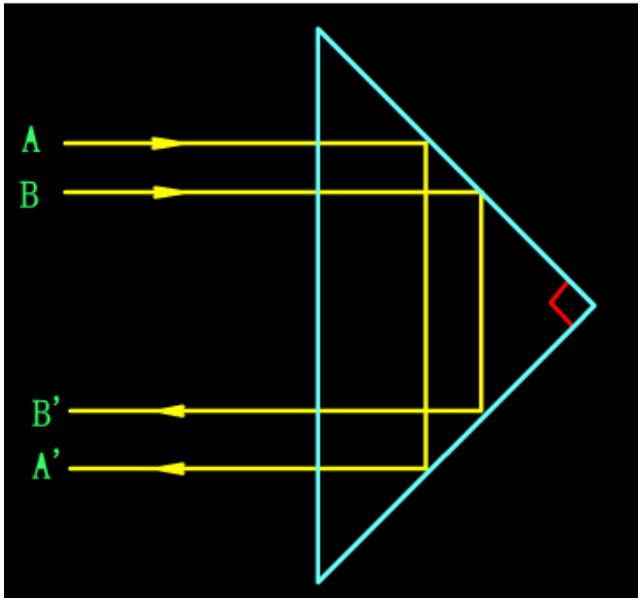


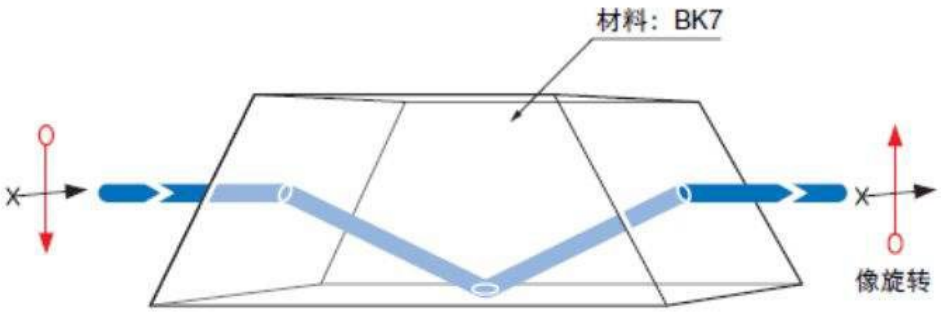
$n_1 > n_2$

全反射现象的应用 —— 棱镜

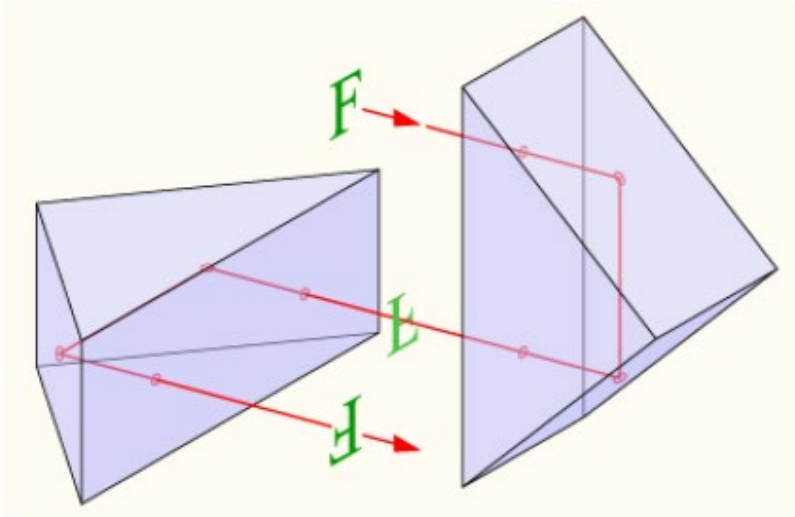








全反射现象的应用 —— 棱镜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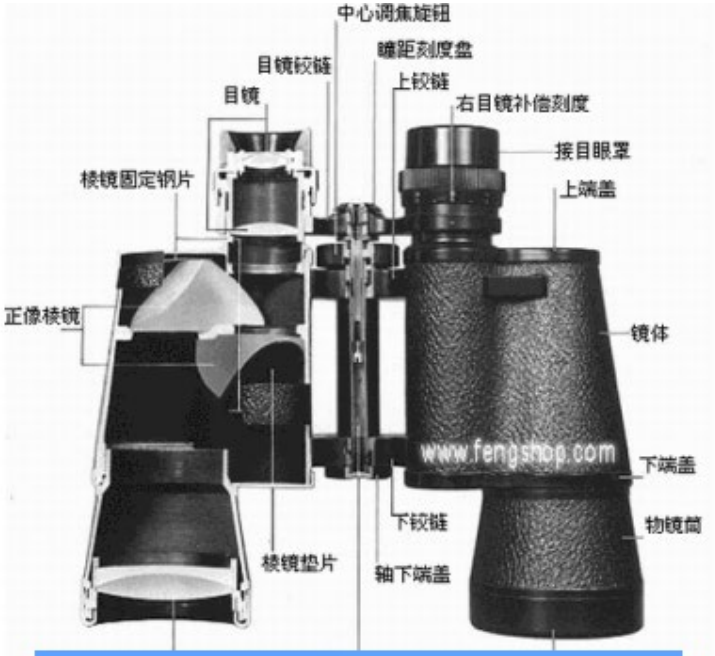
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瞳距刻度盘
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右目镜补偿刻度
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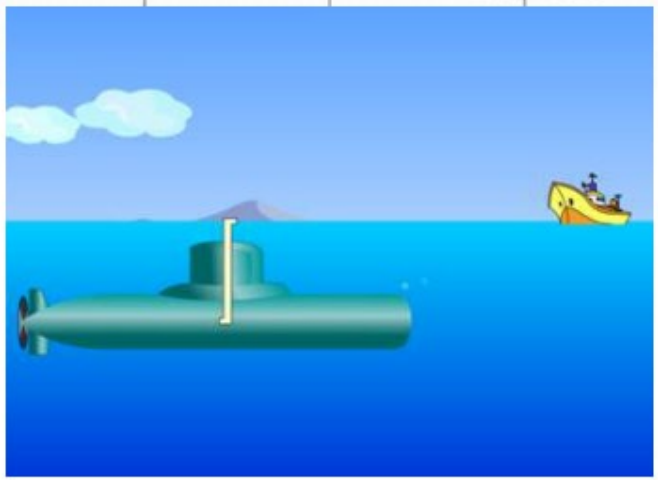
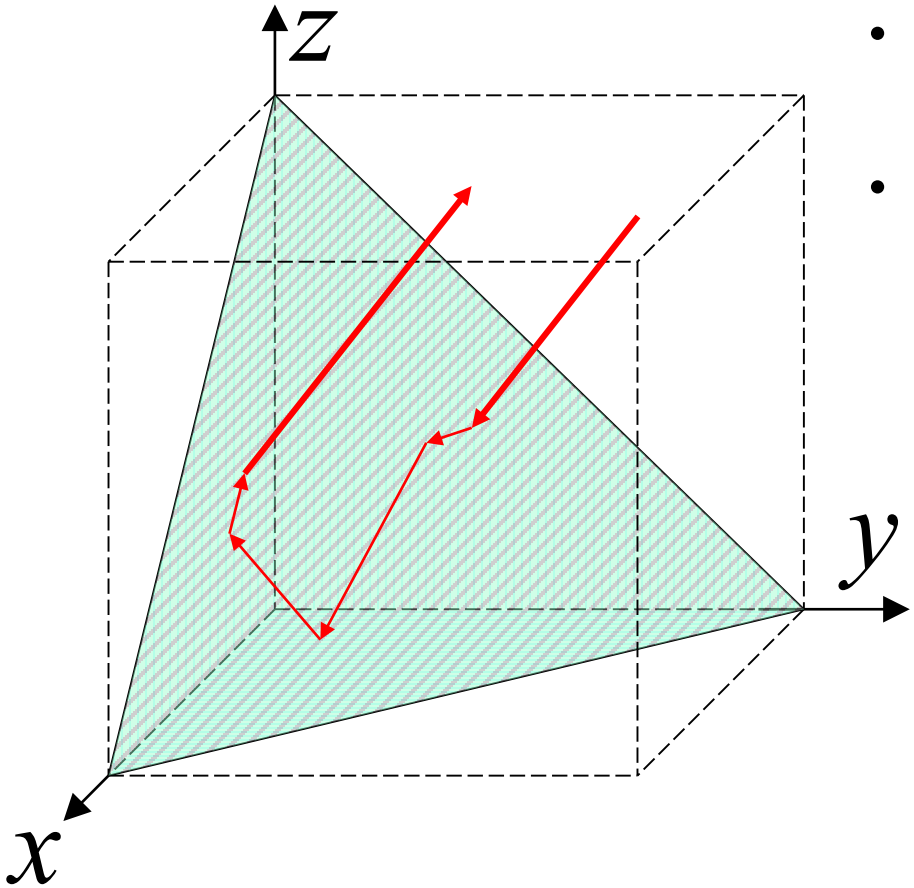
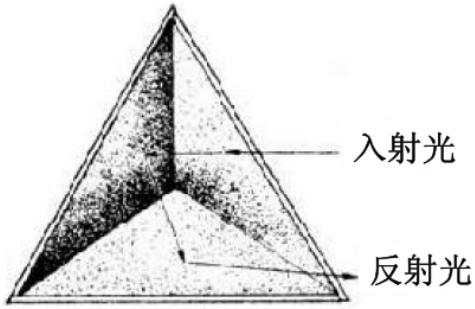
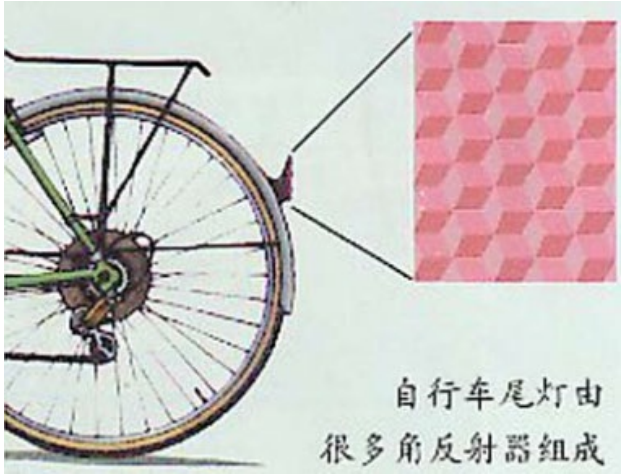


全反射现象的应用 —— 棱镜 ——棱锥反射体



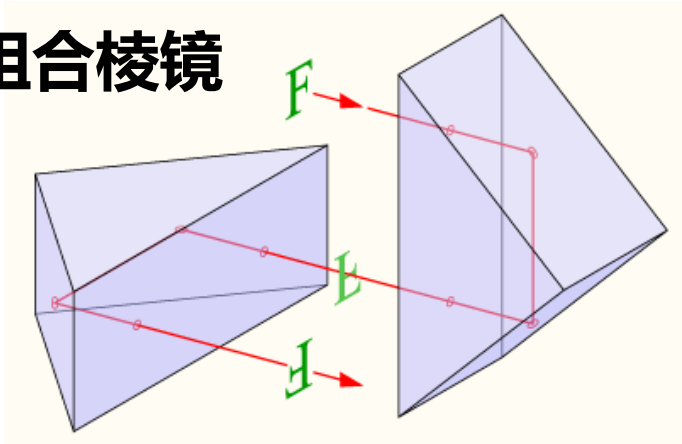
- 从斜面射入、再经三个直角面全反射、最后从斜面出射的光，沿原路返回
- 棱锥阵列反射器：阿波罗棱镜阵列、自行车尾灯、路牌、……



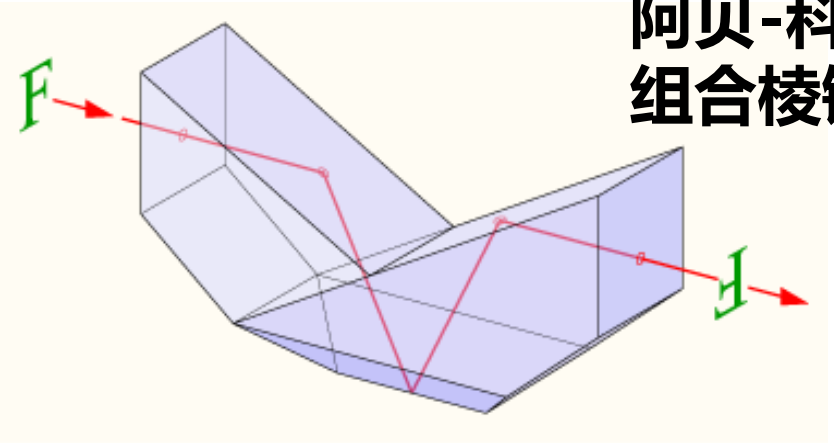


全反射现象的应用 —— 全反射棱镜组合

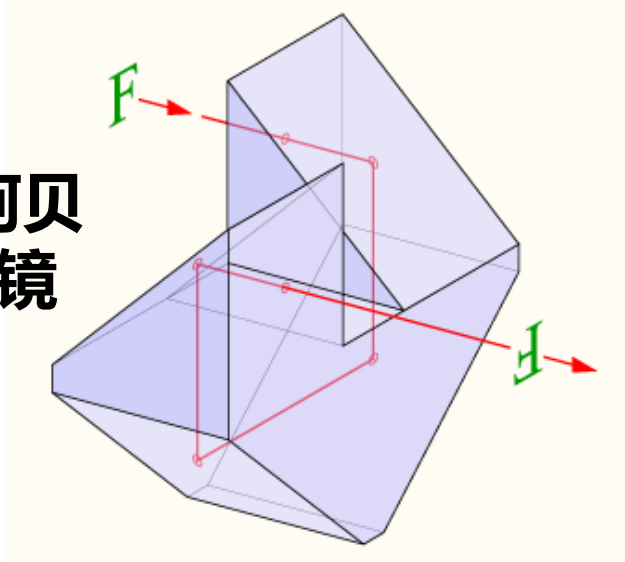
珀罗组合棱镜



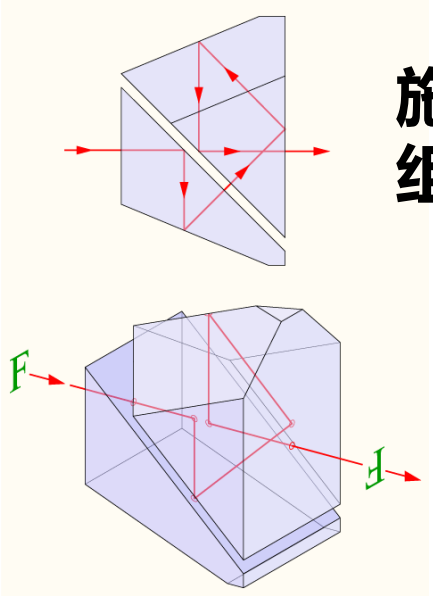
阿贝-科尼组合棱镜



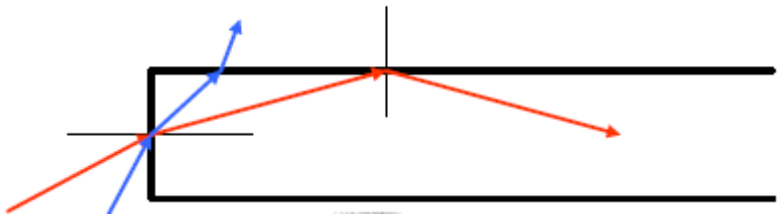
珀罗-阿贝组合棱镜

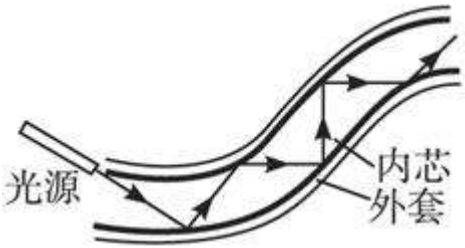


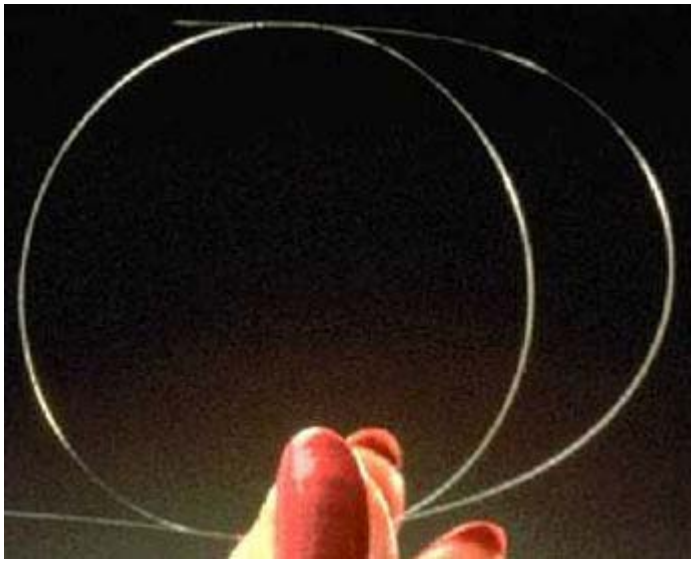
施密特-朴汉组合棱镜



全反射现象的应用 —— 光纤（Optical Fiber）

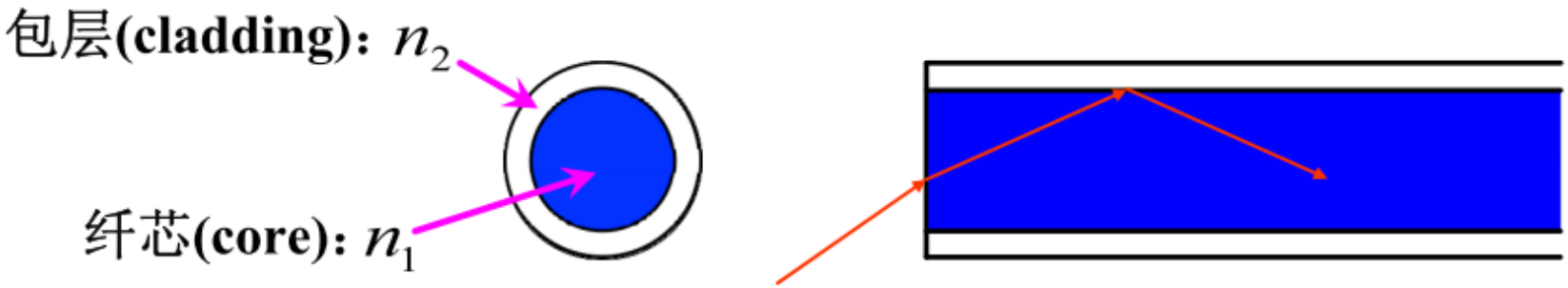




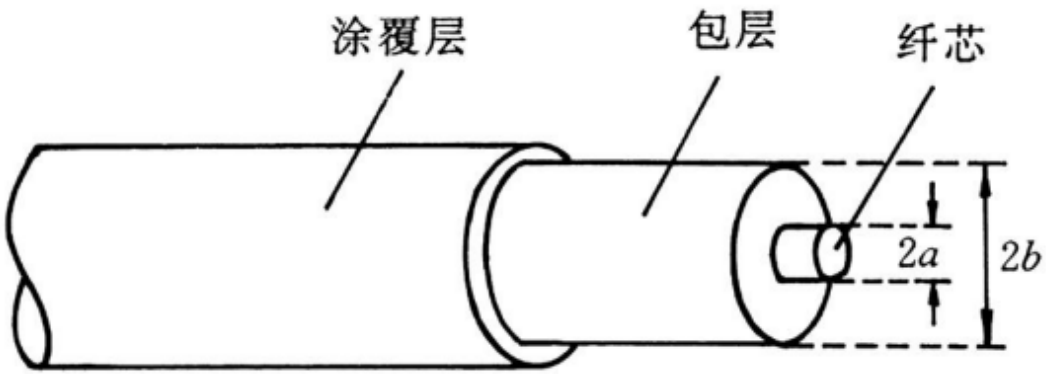


全反射现象的应用 —— 光纤（Optical Fiber）

- 1953，Vanger把一种折射率为1.47的塑料涂在玻璃纤维上，形成比玻璃纤维芯折射率低的套层，得到了光学绝缘的单根纤维。



(step index fiber, SIF)



全反射现象的应用 —— 光纤（Optical Fiber）

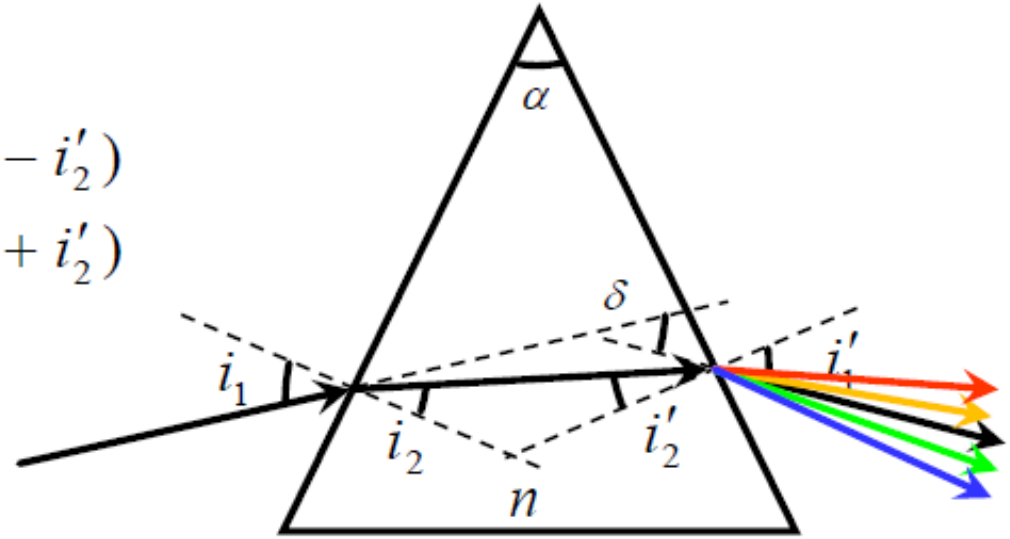
1966，英籍华人高锟(C Kao)指出：如果能够减少玻璃中的杂质含量，就可以制造出损耗低于20dB/km的光纤。

2009 Nobel Laureate, Charles Kao, Father of Fiber Optics

§1.3 棱镜与色散

棱镜的偏向角δ

$$\delta = (i_1 - i_2) + (i_1' - i_2')$$
$$= (i_1 + i_1') - (i_2 + i_2')$$
$$\alpha = i_2 + i_2'$$
$$\delta = (i_1 + i_1') - \alpha$$



最小偏向角：δ_m

$$n = \frac{\sin\frac{\alpha + \delta_m}{2}}{\sin\frac{\alpha}{2}}$$

此时：$i_1 = i_1'$，$i_2 = i_2'$

提供了一种测量介质折射率的方法

证明：

$$\frac{\mathrm{d}\delta}{\mathrm{d}i_1}=1+\frac{\mathrm{d}i_1'}{\mathrm{d}i_1}.$$

产生最小偏向角的必要条件是

$$\frac{\mathrm{d}\delta}{\mathrm{d}i_1}=0 \quad 即 \quad \frac{\mathrm{d}i_1'}{\mathrm{d}i_1}=-1$$

按折射定律

$$\begin{cases} n\sin i_2=\sin i_1, \\ n\sin i_2'=\sin i_1'. \end{cases}$$

取微分后得

$$\begin{cases} n\cos i_2\,\mathrm{d}i_2=\cos i_1\,\mathrm{d}i_1, \\ n\cos i_2'\,\mathrm{d}i_2'=\cos i_1'\,\mathrm{d}i_1'. \end{cases}$$

由上述两式得

$$\frac{\mathrm{d}i_1'}{\mathrm{d}i_1}=\frac{\cos i_1\cos i_2'}{\cos i_2\cos i_1'}\frac{\mathrm{d}i_2'}{\mathrm{d}i_2}$$

由式(1.8)知，$\mathrm{d}i_2=-\mathrm{d}i_2'$

$$\frac{\mathrm{d}i_1'}{\mathrm{d}i_1}=-\frac{\cos i_1\cos i_2'}{\cos i_2\cos i_1'}$$

产生最小偏向角的条件为

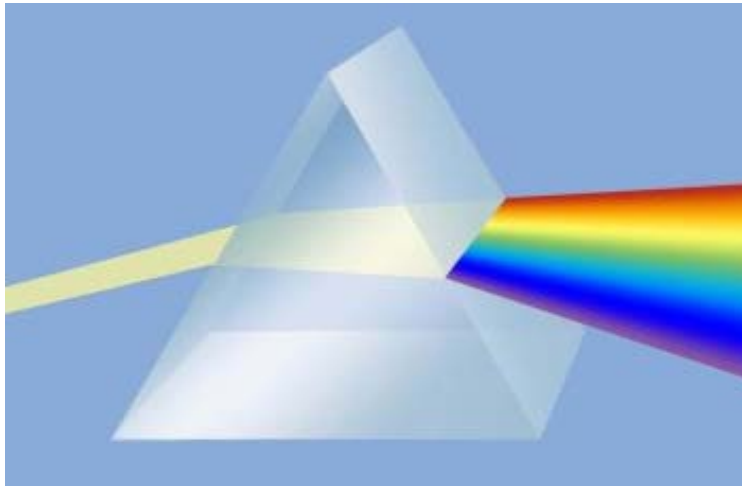
$$\frac{\cos i_1}{\cos i_2}=\frac{\cos i_1'}{\cos i_2'}.$$

平方

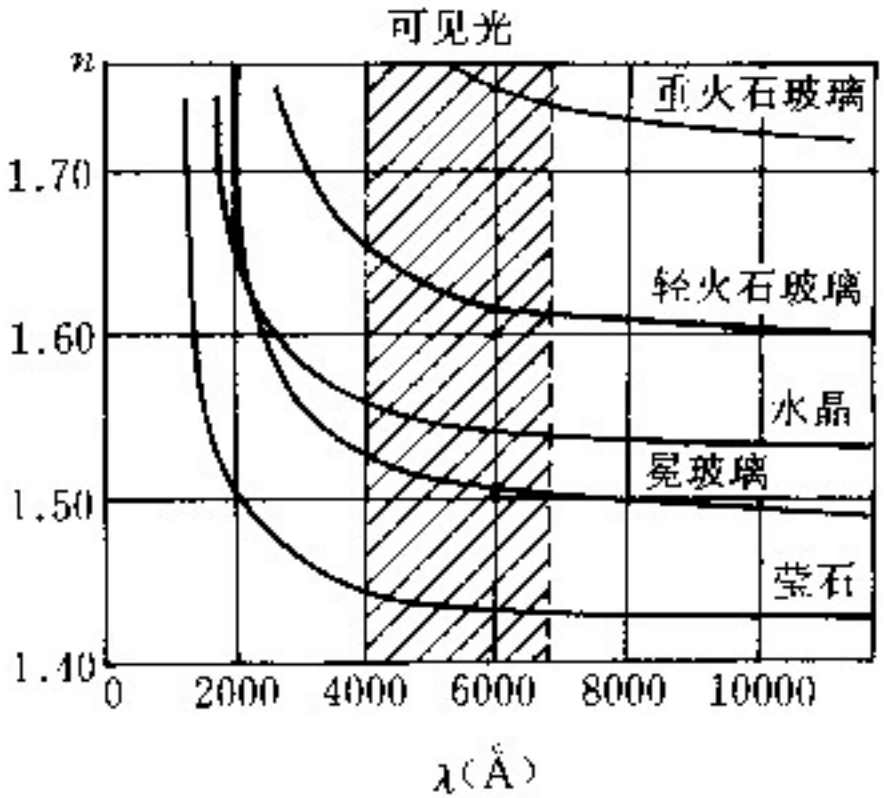
$$\frac{1-\sin^2 i_1}{n^2-\sin^2 i_1}=\frac{1-\sin^2 i_1'}{n^2-\sin^2 i_1'}.$$

$$i_2=i_2'=\alpha/2, \quad i_1=i_1'=(\alpha+\delta_m)/2$$

§1.3 棱镜与色散

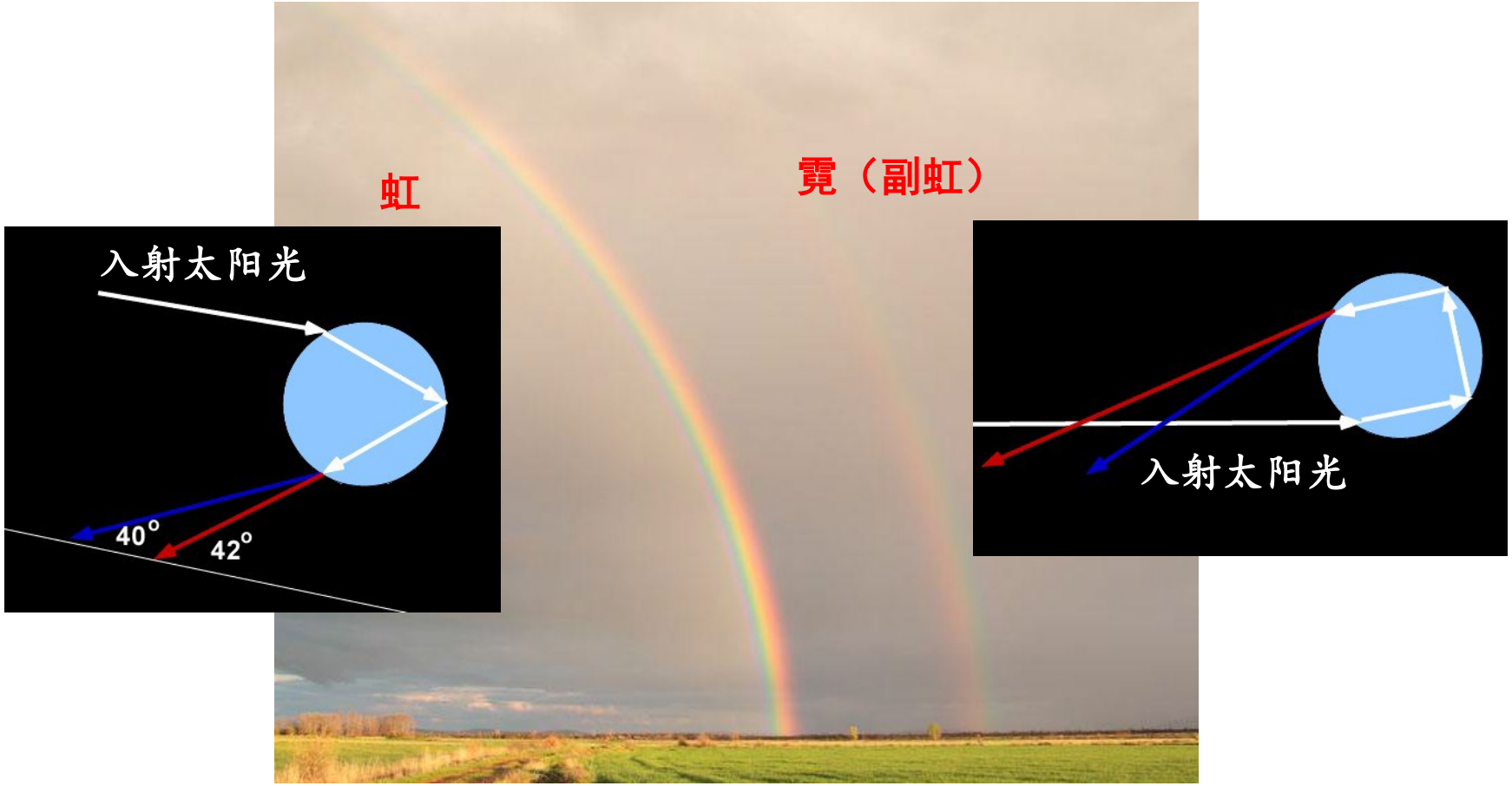


介质折射率n与光的波长有关，这一现象叫做色散。



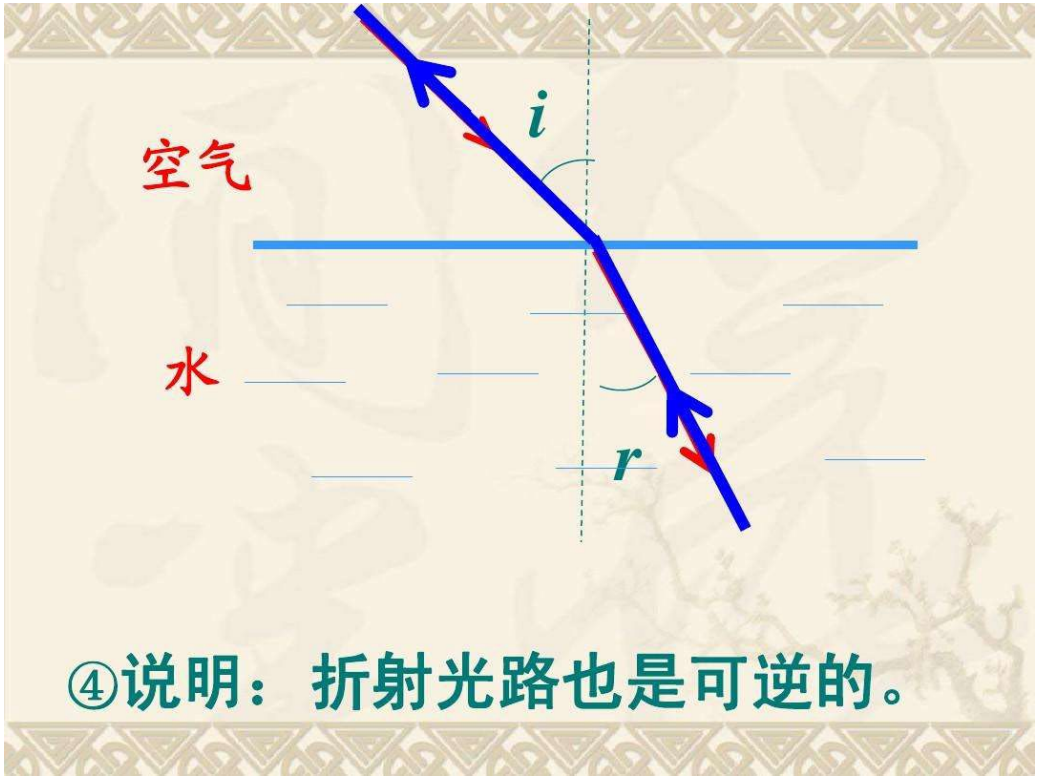
正常色散 $\frac{dn}{d\lambda} < 0$　反常色散 $\frac{dn}{d\lambda} > 0$

虹(Rainbow)和霓(Secondary Rainbow)



§1.4 光的可逆性原理

当光线的方向反转时，它将逆着同一路径传播



作业

- 重排版 P16-17：3, 4, 6, 7, 9, 11, 12